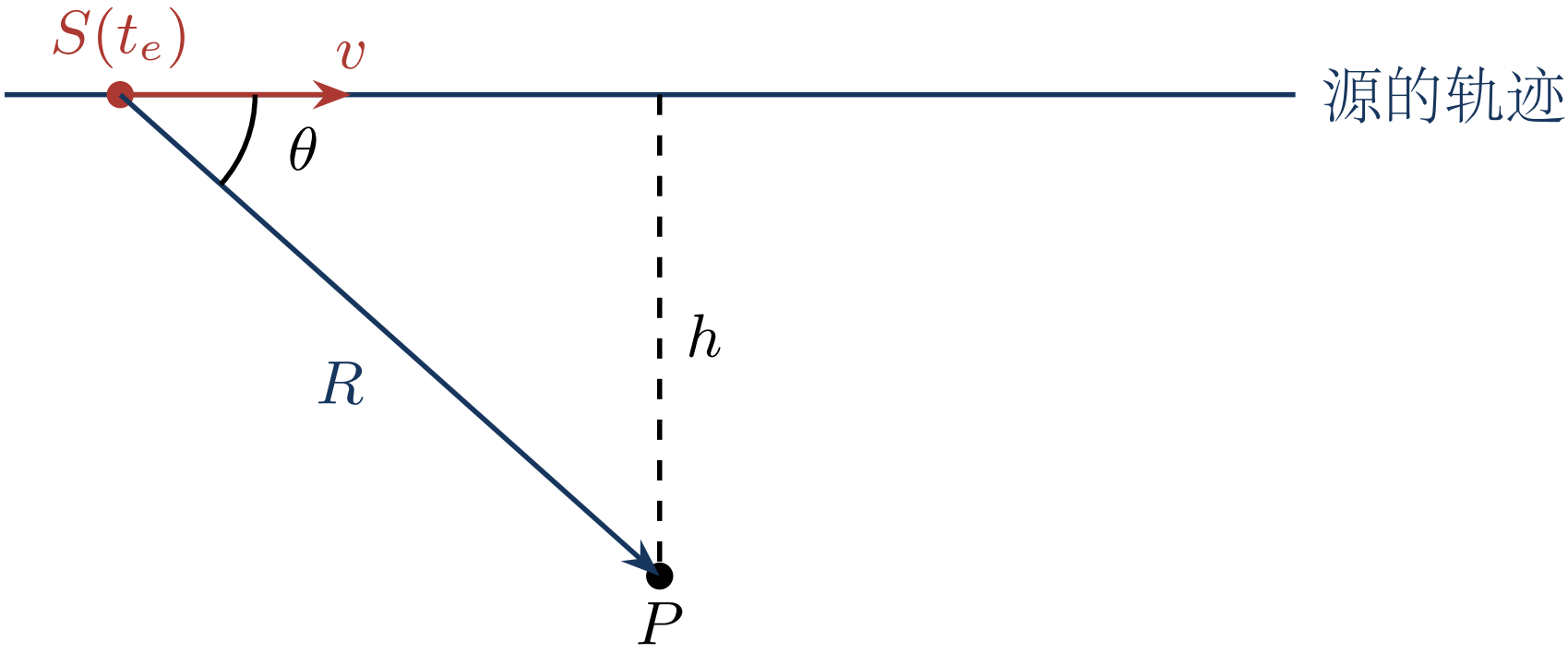

几何角取发射时刻：$\cos\theta = -x/R$。接收时刻 $t_a = t_e + R/u$。